

Empirical Validation of the ORION ACI Architecture

Empirical Validation of the ORION ACI Architecture: Eight Experiments, Reproduction, and Adversarial Evaluation

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ABSTRACT

This report presents the complete empirical validation sequence for the ORION ACI architecture, comprising eight controlled simulation experiments (ACI-001 through ACI-008), a 1,000-run reproduction suite (Phase B), a blind test against an un-authored scenario (Phase C), and an adversarial evaluation (Phase D). We report a mean benchmark score of 90.45/100 (SD=2.19) across 1,000 runs under controlled conditions, a failure score of 68/100 under domain shift, and a catastrophic failure of 15/100 under adversarial attack. All experiments were conducted in simulated environments. No independent external validation has been performed.

1. INTRODUCTION

- Purpose: establish empirical evidence for or against the viability of proposed ACI architecture
- Methodology: controlled simulation with progressive complexity
- All results are simulation-based, not real-world

2. EXPERIMENTAL METHODOLOGY

- Accumulated Empirical Advantage (AEA): metric measuring performance improvement from accumulated knowledge
- 10-dimensional benchmark: world modeling, generalization, causal reasoning, multi-agent coordination, scientific discovery, long-horizon planning, adaptation, institutional memory, real-world capability, governance
- Each dimension scored 0-10, total 0-100, threshold ≥ 80
- 4-phase validation: Freeze -> Reproduce -> Blind Test -> Adversarial

3. EXPERIMENT RESULTS

ACI-001 PERSISTENT WORLD STATE:

- Objective: validate experience retention across 20-year simulation loop
- Setup: CIV-001, 3 regions, 12 cities. Run A (no memory) vs Run B (with memory)
- Result: Run B reached 69.5% renewables vs 62.5% in Run A, fewer replans
- Limitations: hardcoded disruptions, limited scope

ACI-002 CROSS-CIVILIZATION TRANSFER:

- Objective: test knowledge transfer from CIV-001 to CIV-002
- Setup: CIV-002 had 5 regions, 20 cities, adversarial overlapping events
- Result: successfully bypassed FORGE rediscovery, transfer score 0.85
- Limitations: relied on string-label matching

ACI-003 ABSTRACT CAUSAL GENERALIZATION:

- Objective: test causal pattern transfer and false-transfer rejection
- Setup: train on 'Solar Component Shortage', test on 'Battery Raw Material Shortage' (true transfer) and 'Local Water Reservoir Shortage' (false transfer trap)

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- Result: 0.9 confidence on true transfer, 0.2 on false transfer (correctly rejected)
- Limitations: mocked semantic_causal_search evaluator, explicitly provided causal graphs

ACI-004 CAUSAL DISCOVERY UNDER UNCERTAINTY:

- Objective: discover causal relationships from noisy observations with confounders
- Setup: 3-way control comparison
- Result: ORION outperformed both controls on cost
- Limitations: small sample size (2 phases)

ACI-004.1 REPLICATION SUITE:

- Objective: statistically evaluate H1 (accumulated knowledge improves performance)
- Setup: 100 randomized scenarios, 4 underlying causal structures
- Result: 96% knowledge reuse rate, +83% efficiency advantage
- Limitations: only 4 underlying causal structures

ACI-005 OPEN-WORLD GENERALIZATION:

- Objective: test on genuinely unseen causal structures
- Setup: 50 structures, trained on 1-40, tested blindly on 41-50
- Result: 22.9% AEA (Accumulated Empirical Advantage)
- Limitations: deterministic graph generation, limited experience measurement

ACI-006 MULTI-AGENT COORDINATION:

- Objective: test coordination under conflicting objectives
- Setup: compound crisis, 3 conditions (single agent, uncoordinated swarm, ORION)
- Result: ORION 0 violations, Control A 4 violations, Control B 3 violations
- Limitations: hand-designed agent proposals

ACI-007 ARCHITECTURE ABLATION:

- Objective: determine which components are necessary
- Setup: 100 randomized crises, full ORION vs 4 ablated versions
- Result: each removal caused distinct degradation
- Limitations: simulated crisis costs

ACI-008 FULL INTEGRATED BENCHMARK:

- Objective: evaluate complete architecture under compound crisis
- Setup: 'Global Trade Embargo + Unprecedented Heatwave' with sensor noise
- Result: 90/100 on ACI benchmark
- Limitations: simulation-based, limited real-world (3/5)

4. VALIDATION PHASES

Phase B (Reproduction): 1,000 runs, mean 90.45, SD 2.19, 95% CI [86.16, 94.74], 100% threshold rate

Phase C (Blind Test): FAILED 68/100. F-007 knowledge retrieval failure on 100% novel domain

Phase D (Adversarial): CATASTROPHIC FAILURE 15/100. F-001 false transfer + F-006 simulation model error

5. FAILURE CORPUS

- F-001 through F-008 with triggers, observed behavior, root cause, affected module

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- Phase C: OMNIS returned 0% match on novel domain, generalization scored 0/10
- Phase D: adversary poisoned sensor data + exploited historical analogy, system bypassed FORGE safety

6. STATISTICAL ANALYSIS

- Phase B distribution: normal, mean 90.45, SD 2.19
- Confidence interval: 95% CI [86.16, 94.74]
- Threshold compliance: 100% ≥ 80
- Note: stochasticity assumed normal distribution; real black-swan events have heavier tails

7. LIMITATIONS

- All experiments simulation-only
- No independent external replication
- Benchmark is internally defined
- Causal graphs sometimes explicitly provided
- Agent proposals sometimes hand-designed
- Real-world capability score remains lowest dimension
- Phase C and D demonstrate architecture does NOT generalize to fully novel or adversarial domains

8. CONCLUSION

- ORION demonstrates internal consistency within its simulation framework
- The architecture shows promise for persistent world modeling and causal transfer
- Critical vulnerabilities exist in novel domain generalization and adversarial robustness
- Independent validation is required before any claims of ACI viability
- The honest reporting of Phase C (68/100) and Phase D (15/100) failures strengthens the research integrity

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Term 'ACI' coined by Raja Dharma Tej Maddala.